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Growing Homes: Manipulating Morphogenesis to Shape Houses from Seeds

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Problem Statement: Traditional construction methods rely heavily on non-renewable resources, generate significant waste, and contribute to environmental degradation. As urbanization increases, alternative and sustainable construction techniques are needed to reduce the ecological footprint of human habitation.

Objective: This research explores the potential of using morphogenesis—the biological process that guides organismal shape formation—to grow houses from seeds. By manipulating plant growth through genetic engineering and controlled environmental conditions, we can develop living structures that are self-sustaining, biodegradable, and environmentally friendly.

What is Morphogenesis? Morphogenesis is the biological process by which cells organize into complex structures, determining an organism's shape and function. This principle can be applied to plants to influence their growth patterns, allowing us to shape natural structures for habitation.

Current Trends in Bio-Architecture: Living architecture is an emerging field that integrates biological processes into building design. Advances in biodesign have led to plant-based materials, fungal bricks, and algae-powered energy solutions. Researchers are exploring self-growing structures that adapt to environmental conditions.[2]

This is a concept of what a grown house could look like. I uses chat gpt to generate these images by asking it to create an image that is a futuristic organic house grown from a seed, seamlessly blending nature and architecture with living walls, bioluminescent leaves, and intertwining roots forming a self-sustaining structure.

Integrating living plants into architectural design offers a sustainable approach to construction, blending nature and technology to create adaptive, eco-friendly structures.[1]

Summary: The application of morphogenesis to architecture presents an innovative approach to sustainable construction. By harnessing the principles of biology, we can create living, self-sustaining homes that reduce environmental harm and adapt to ecological changes.

Call to Action: Further interdisciplinary collaboration is necessary to refine techniques, develop hybrid bio-engineering methods, and bring bio-architectural innovations to practical implementation.

references

[1] Charlois, "Living Plants in Architecture and Design," *Charlois*, [Online]. Available: https://charlois.com/en/living-plants-in-architecture-and-design/. [Accessed: 09-Feb-2025].

[2] M. Jones, A. Mautner, S. Luenco, A. Bismarck, and S. John, "Engineered mycelium composite construction materials from fungal biorefineries: A critical review," *Materials & Design*, vol. 187, pp. 108397, 2020. [Online]. Available: https://doi.org/10.1016/j.matdes.2019.108397. [Accessed: 09-Feb-2025].